

A Technical Introduction to Ensemble learning

Section 1

$$e^k = H(p, q^k) - \frac{\lambda}{K} \sum_{j \neq k} H(q^j, q^k) \quad \{\displaystyle e^k = H(p, q^k) - \frac{\lambda}{K} \sum_{j \neq k} H(q^j, q^k)\}$$

Section 2

The Bayes optimal classifier is a classification technique. It is an ensemble of all the hypotheses in the hypothesis space. On average, no other ensemble can outperform it. The Naive Bayes classifier is a version of this that assumes that the data is conditionally independent on the class and makes the computation more feasible. Each hypothesis is given a vote proportional to the likelihood that the training dataset would be sampled from a system if that hypothesis were true. To facilitate training data of finite size, the vote of each hypothesis is also multiplied by the prior probability of that hypothesis. The Bayes optimal classifier can be expressed with the following equation:

Section 3

$$y = \arg \max_{c_j \in C} \sum_{h_i \in H} P(c_j | h_i) P(T | h_i) P(h_i) \quad \{\displaystyle y = \underset{c_j \in C}{\operatorname{argmax}} \sum_{h_i \in H} P(c_j | h_i) P(T | h_i) P(h_i)\}$$

Section 4

Ensemble learning applications In recent years, due to growing computational power, which allows for training in large ensemble learning in a reasonable time frame, the number of ensemble learning applications has grown increasingly. Some of the applications of ensemble classifiers include:

Section 5

Bayesian model combination Bayesian model combination (BMC) is an algorithmic correction to Bayesian model averaging (BMA). Instead of sampling each model in the ensemble individually, it samples from the space of possible ensembles (with model weights drawn randomly from a Dirichlet distribution having uniform parameters). This modification overcomes the tendency of BMA to converge toward giving all the weight to a single model. Although BMC is somewhat more computationally expensive than BMA, it tends to yield dramatically better results. BMC has been shown to be better on average (with statistical significance) than BMA and bagging. Use of Bayes' law to compute model weights requires computing the probability of the data given each model. Typically, none of the models in the ensemble are exactly the distribution from which the training data were generated, so all of them correctly receive a value close to zero for this term. This would work well if the ensemble were big

enough to sample the entire model-space, but this is rarely possible. Consequently, each pattern in the training data will cause the ensemble weight to shift toward the model in the ensemble that is closest to the distribution of the training data. It essentially reduces to an unnecessarily complex method for doing model selection. The possible weightings for an ensemble can be visualized as lying on a simplex. At each vertex of the simplex, all of the weight is given to a single model in the ensemble. BMA converges toward the vertex that is closest to the distribution of the training data. By contrast, BMC converges toward the point where this distribution projects onto the simplex. In other words, instead of selecting the one model that is closest to the generating distribution, it seeks the combination of models that is closest to the generating distribution. The results from BMA can often be approximated by using cross-validation to select the best model from a bucket of models. Likewise, the results from BMC may be approximated by using cross-validation to select the best ensemble combination from a random sampling of possible weightings.

Section 6

Distributed denial of service Distributed denial of service is one of the most threatening cyber-attacks that may happen to an internet service provider. By combining the output of single classifiers, ensemble classifiers reduce the total error of detecting and discriminating such attacks from legitimate flash crowds.

Section 7

While speech recognition is mainly based on deep learning because most of the industry players in this field like Google, Microsoft and IBM reveal that the core technology of their speech recognition is based on this approach, speech-based emotion recognition can also have a satisfactory performance with ensemble learning. It is also being successfully used in facial emotion recognition.

Section 8

Land cover mapping Land cover mapping is one of the major applications of Earth observation satellite sensors, using remote sensing and geospatial data, to identify the materials and objects which are located on the surface of target areas. Generally, the classes of target materials include roads, buildings, rivers, lakes, and vegetation. Some different ensemble learning approaches based on artificial neural networks, kernel principal component analysis (KPCA), decision trees with boosting, random forest and automatic design of multiple classifier systems, are proposed to efficiently identify land cover objects.

Section 9

For each model m in the bucket: Do c times: (where ' c ' is some constant) Randomly divide the training dataset into

two sets: A and B Train m with A Test m with B Select the model that obtains the highest average score

Section 10

Bucket of models A "bucket of models" is an ensemble technique in which a model selection algorithm is used to choose the best model for each problem. When tested with only one problem, a bucket of models can produce no better results than the best model in the set, but when evaluated across many problems, it will typically produce much better results, on average, than any model in the set. The most common approach used for model-selection is cross-validation selection (sometimes called a "bake-off contest"). It is described with the following pseudo-code:

Section 11

where e_k is the cost function of the k classifier, q_k is the probability of the k classifier, p is the true probability that we need to estimate and λ is a parameter between 0 and 1 that define the diversity that we would like to establish. When $\lambda = 0$ we want each classifier to do its best regardless of the ensemble and when $\lambda = 1$ we would like the classifier to be as diverse as possible.